

Ornamental letters images registration and pre-processing

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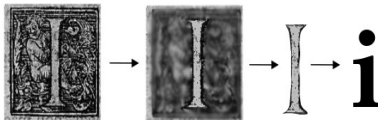
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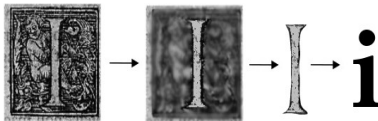
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Introduction

An interesting problem in the CALYPOD project : finding the letter represented on a given image $\longrightarrow$ **segmentation** of the image.



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To help segmentation :



Image registration (rotation,
translation, scale)



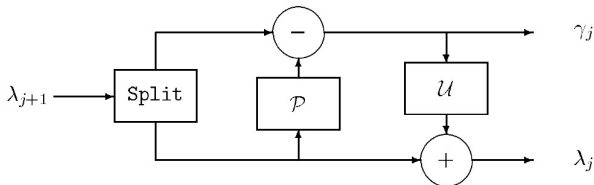
Image pre-processing
(cleaning)

Image pre-processing

- Second generation wavelets [1],
- Multiresolution analysis using biorthogonal wavelets,
- Three steps :
 - Split,
 - Predict,
 - Update.

The lifting scheme

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- Multiresolution analysis using biorthogonal wavelets,
- Three steps :
 - Split,
 - Predict,
 - Update.
- λ_{j+1} is the original signal, λ_j is the lower resolution signal and γ_j represents the lost details when passing from λ_{j+1} to λ_j .



Integers to integers wavelet transform

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- Integer to integer wavelet coefficients [2],
- Fast and easy computation (integer images),
- Multiresolution analysis.

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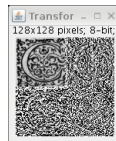
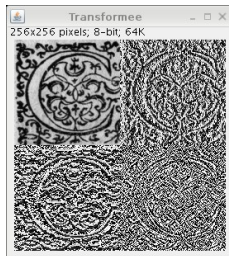
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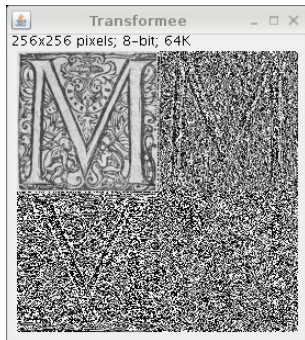
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- Fast and easy computation (integer images),
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Wavelets give :

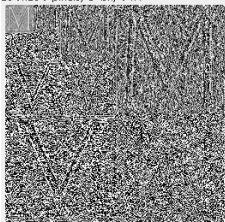
- Approximation of the original image $\rightarrow$ shape of the letter (global view).
- Details (horizontal, vertical and diagonal) images $\rightarrow$ texture (local view).



To detect the shape of the letter :

- ① Compute lifting scheme at several resolution levels,
- ② Put details to zero,
- ③ Reconstruct original image without details.

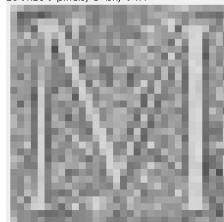
256x256 pixels; 8-bit; 64K



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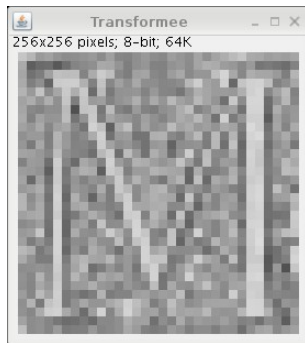


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To detect the shape of the letter :

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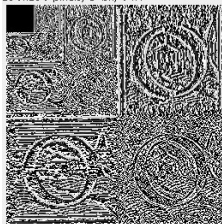
To work with details : work directly with horizontal, vertical and diagonal details, **OR** :

- ① Compute lifting scheme at several resolution levels,
- ② Put approximation to zero,
- ③ Reconstruct original image without approximation.

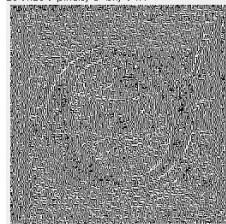
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In our work we propose a method based on the lifting scheme to allow pre-processing of images $\rightarrow$ shape recognition.

TODO :

- Check the proposed transform on many ornamental letters,
- Find the more accurate wavelet family for the problem,
- Improve the reconstructed letter,
- Build a set of test images,
- Apply segmentation algorithms (level-sets),
- Test recognition using Support Vector Machine classifier.

Image registration

- Classical approach : cross-correlation (CC) between image (i) and reference (r).

$$CC(x, y) = \sum_{x', y'} i(x, y) r(x - x', y - y'). \quad (1)$$

$\Rightarrow$ peak coordinates = translation vector.

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- Noise sensitivity.
- Peak dynamic is low.

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- Computation in Fourier domain.
- Reference and patterns in the image must be identical.
- Noise sensitivity.
- Peak dynamic is low.
- $\Rightarrow$ Keep only phase information $\Rightarrow$ SPOMF algorithm (Symetric Phase Only Matched Filters).

- SPOMF algorithm [3] (in Fourier domain) :

$$Q(u, v) = \frac{I(u, v)R(u, v)^*}{|I(u, v)| |R(u, v)|} \quad (2)$$

(Q , I and R are the Fourier transforms of the images).

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- Noise robustness
- Allow differences between reference and patterns in the image.
- $\Rightarrow$ Registration in translation OK.
- $\Rightarrow$ Registration in rotation/scaling via Fourier Mellin invariants (FMI).

Some Fourier transformation properties :

- modulus is invariant to translation,
- rotate image i by $\theta \Rightarrow I$ is also rotated by θ ,
- scale image i by $\alpha \Rightarrow I$ is scaled by $1/\alpha$.

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- scale image i by $\alpha \Rightarrow I$ is scaled by $1/\alpha$.

Key idea : transform rotation and scaling in translations (to be detectable via SPOMF).

- rotation $\Rightarrow$ translation : rectangular $\Rightarrow$ polar coordinates.
- scaling $\Rightarrow$ translation : logarithmic coordinates.

FMI-SPOMF registration algorithm

Given image i and reference r :

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- compute I and R , Fourier transforms of i and r ,
- apply a log-polar coordinates transformation to $|I|$ and $|R|$ (involve interpolation) :

$$x(\rho, \theta) = \frac{W}{\sqrt{2}} K^{(\rho-R)} \sin\left(\theta \frac{\pi}{T}\right) \quad (3)$$

$$y(\rho, \theta) = \frac{H}{2} + \frac{W}{\sqrt{2}} K^{(\rho-R)} \cos\left(\theta \frac{\pi}{T}\right) \quad (4)$$

(W, H) is image i size, (R, T) transformed-image size, K is a parameter given highest scaling registrable.

FMI-SPOMF registration algorithm(2)

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- apply SPOMF between these images, giving (x, y) as translation vector :

$$scale = K^x \quad (5)$$

$$rotation = y \frac{\pi}{T}, \quad (6)$$

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- the translation registration is then obtained by using SPOMF on the rotation/scaling registered image.

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- **TODO** :
 - finalize implementation,
 - test with larger amount of ornamental letters.

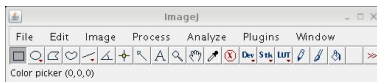
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- Tools to study ornamental letters are being developed in order to achieve :

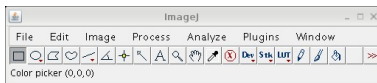
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 - Image registration to ensure good image orientation,
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- A lot of work to do on the CALYPOD project but shared tools will help us to reach the goals of the project.



W. Sweldens.

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SIAM J. Math. Anal., 29(2) :511–546, 1997.



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Thanks for your attention...